

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458036.76132 +/- 0.00088 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1591 +/- 0.0027
Transit depth (Rp/Rs)^2	2.53 +/- 0.085 [%]
Orbital Inclination (inc)	89.94 +/- 0.52
Airmass coefficient 1 (a1)	1.1688 +/- 0.0075
Airmass coefficient 2 (a2)	-0.1439 +/- 0.0029
Scatter in the residuals of the lightcurve fit is	0.65 %
Best Comparison Star	None
Optimal Aperture	4.56
Optimal Annulus	6.83
Transit Duration (day)	0.13177 +/- 0.00037

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.1591 $\pm$ 0.0027 vs 0.1489 (deviation 2.13 σ)
Duration fit vs published	0.13177 d vs 0.1304 d (deviation 2.06 σ)
Mid-time O-C	0.8 min
Depth SNR	33.2
Residual scatter	0.65 %

Light curve

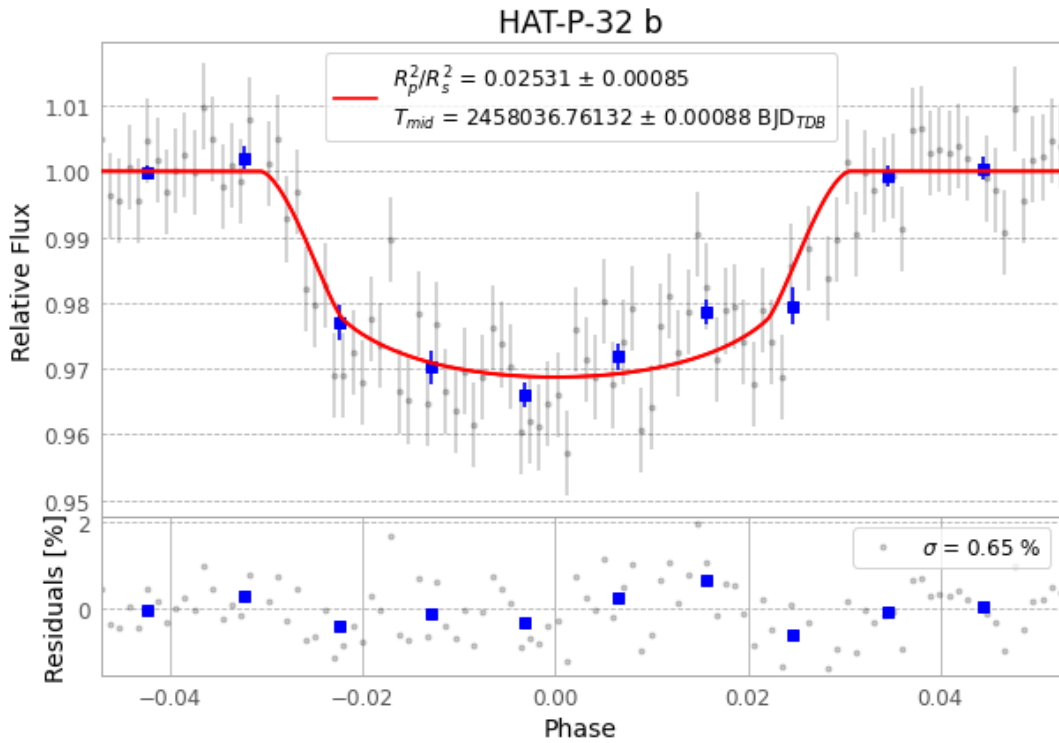


Figure 1 — FinalLightCurve_HAT-P-32 b_2017-10-09.png (SHA-256 480fbfe48aebc1bf...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [432, 263], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 01a8d6f33e322af4...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13

Data provenance

104 FITS frames (68 MB), HATP-32171010034212.FITS → HATP-32171010085112.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-32-171010.log (93cbe69181f4...), FinalLightCurve_HAT-P-32 b_2017-10-09.png (480fbfe48aeb...), FinalLightCurve_HAT-P-32 b_2017-10-09.csv (230c1853bccc...)

Compute resources

Wall time 148.0 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)